

MadhuTwin architecture: two data streams fused into a living digital twin

Type 2 Diabetes · predicts glucose spikes and hypoglycaemia up to 2 hours ahead · doctor-facing dashboard

1 · Data streams

Static / historical: EHR

Demographics, diagnoses (SNOMED CT)
Labs: HbA1c, FPG, insulin, lipids, eGFR (LOINC)
Medications with timing (WHO ATC)
Family history · TCF7L2 rs7903146 · polygenic risk
FHIR R4 bundles, ABDM-aligned

Dynamic / real-time: wearables & IoT

CGM every 5–15 min
Heart rate, HRV (RMSSD), steps, METs
Sleep stages (hypnogram)
Logged meals (Indian foods) and doses

Sources

Synthetic India cohort 1,000 × 14 d · CGMacros (45, real) · ShanghaiT2DM (100, external)

2 · Digital twin core

Ingest & fuse

Common 5-min schema · missing-aware · EHR vector + 21 dynamic channels

MODEL

Personalised glucose–insulin physiology (MAP fit, EHR priors)

SYNC

Unscented Kalman filter every 5 min: state + insulin-sensitivity drift

SIMULATE

2-h forecasts and what-if: meals, walks, sleep, illness

TwinNet: physics-gated fusion network

Causal dilated TCN over 6 h × 21 channels
EHR encoder conditions the dynamics (FiLM)
Gate learns how much to trust the physics twin per horizon
Outputs: 5–120 min quantiles (conformal 80% band) + spike / hypo probability
LightGBM + SHAP: plain-language reasons

3 · Clinician experience

FastAPI + WebSocket

Live replay · audit trail · FHIR export

Doctor dashboard

Risk-ranked remote-monitoring panel
Virtual patient: organ-level physiology
24 h CGM + 2 h forecast cone + alerts with reasons
AGP report (international consensus)
What-if simulator with Indian meals
Ask the twin (grounded assistant)

Interoperability

FHIR R4 RiskAssessment carries the twin's forecast to any EHR